

## HA'APAI, Tonga

WMO: 917840

Lat: 19.777S

Lon: 174.341W

Elev: 9

StdP: 101.21

Time Zone: 13.00 (E13)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 8             | 19.2       | 20.1 | 12.7                          | 9.2 | 21.5 | 13.6 | 9.7 | 21.9 | 10.5                  | 23.3 | 10.0 | 23.0 | 2.0                   | 180  | 0.372 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 3.9                    | 30.2            | 26.7 | 30.0 | 26.6 | 29.6 | 26.2 | 27.6                | 29.4 | 27.2 | 29.0 | 26.8 | 28.7 | 3.8                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.1                            | 22.8 | 29.1 | 26.6 | 22.2 | 28.7 | 26.2 | 21.6 | 28.3 | 88.3          | 29.6 | 86.0 | 29.3 | 84.4 | 29.1 | 29.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 10.0          | 9.0  | 8.2 | DB                         | 17.7 | 31.5               | 0.7 | 0.6                                                | 17.2 | 31.9       | 16.7 | 32.2       | 16.3 | 32.5       | 15.8 | 32.9 | (4) |
| (5)               |      |     | WB                         | 14.6 | 28.7               | 0.8 | 0.4                                                | 14.0 | 29.0       | 13.6 | 29.2       | 13.2 | 29.5       | 12.6 | 29.8 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 25.5        | 27.1  | 27.4  | 27.3  | 26.5  | 25.4  | 24.4  | 23.7  | 23.5  | 23.7  | 24.7  | 25.7  | 26.7  | (6)  |  |
|                                                                        | DBStd       | 1.86        | 1.06  | 1.01  | 0.91  | 1.00  | 1.12  | 1.35  | 1.30  | 1.26  | 1.53  | 1.30  | 1.21  | 1.05  | (7)  |  |
|                                                                        | HDD10.0     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |  |
|                                                                        | HDD18.3     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (9)  |  |
|                                                                        | CDD10.0     | 5655        | 531   | 488   | 537   | 496   | 477   | 433   | 423   | 418   | 410   | 454   | 471   | 517   | (10) |  |
|                                                                        | CDD18.3     | 2613        | 272   | 254   | 278   | 246   | 219   | 183   | 165   | 159   | 160   | 196   | 221   | 258   | (11) |  |
|                                                                        | CDH23.3     | 19408       | 2742  | 2651  | 2861  | 2209  | 1444  | 888   | 550   | 459   | 604   | 1042  | 1626  | 2333  | (12) |  |
|                                                                        | CDH26.7     | 3037        | 612   | 657   | 645   | 328   | 91    | 34    | 12    | 11    | 18    | 57    | 173   | 399   | (13) |  |
| (14) Wind                                                              | WSAvg       | 4.2         | 4.0   | 3.8   | 3.6   | 4.1   | 4.5   | 4.1   | 4.2   | 4.4   | 4.2   | 4.5   | 4.3   | 4.5   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1680        | 170   | 194   | 254   | 176   | 106   | 74    | 92    | 101   | 105   | 97    | 122   | 153   | (15) |  |
|                                                                        | PrecMax     | 2661        | 442   | 486   | 592   | 497   | 430   | 231   | 231   | 341   | 294   | 339   | 511   | 613   | (16) |  |
|                                                                        | PrecMin     | 826         | 10    | 50    | 52    | 2     | 0     | 8     | 2     | 12    | 2     | 6     | 3     | 8     | (17) |  |
|                                                                        | PrecStd     | 441         | 109   | 109   | 133   | 116   | 86    | 53    | 64    | 73    | 77    | 79    | 112   | 124   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 30.8  | 31.1  | 30.9  | 30.1  | 29.1  | 28.2  | 27.9  | 27.8  | 28.0  | 28.4  | 29.5  | 30.2  | (19) |  |
|                                                                        |             | MCWB        | 27.2  | 27.4  | 27.4  | 26.2  | 25.7  | 25.1  | 24.8  | 24.6  | 24.7  | 24.9  | 25.7  | 26.0  | (20) |  |
|                                                                        | 2%          | DB          | 30.1  | 30.2  | 30.1  | 29.3  | 28.2  | 27.4  | 27.0  | 26.9  | 27.1  | 27.9  | 28.9  | 29.7  | (21) |  |
|                                                                        |             | MCWB        | 26.8  | 26.8  | 26.7  | 25.7  | 24.9  | 24.5  | 24.0  | 24.0  | 24.0  | 24.5  | 25.2  | 25.8  | (22) |  |
|                                                                        | 5%          | DB          | 29.7  | 30.0  | 29.8  | 29.0  | 27.8  | 27.0  | 26.2  | 26.2  | 26.6  | 27.2  | 28.2  | 29.1  | (23) |  |
|                                                                        |             | MCWB        | 26.6  | 26.6  | 26.4  | 25.4  | 24.5  | 24.2  | 23.3  | 23.3  | 23.4  | 24.0  | 24.6  | 25.6  | (24) |  |
|                                                                        | 10%         | DB          | 29.1  | 29.4  | 29.2  | 28.3  | 27.1  | 26.3  | 25.8  | 25.5  | 25.9  | 26.8  | 27.8  | 28.6  | (25) |  |
|                                                                        |             | MCWB        | 26.1  | 26.2  | 26.0  | 25.0  | 24.0  | 23.7  | 22.8  | 22.6  | 22.8  | 23.8  | 24.4  | 25.3  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 27.8  | 28.4  | 28.0  | 27.2  | 26.5  | 26.2  | 25.7  | 25.4  | 25.6  | 26.1  | 26.6  | 27.3  | (27) |  |
|                                                                        |             | MCDB        | 29.6  | 30.2  | 29.9  | 28.8  | 27.9  | 27.3  | 27.2  | 26.7  | 27.1  | 27.4  | 28.4  | 29.0  | (28) |  |
|                                                                        | 2%          | WB          | 27.4  | 27.6  | 27.4  | 26.6  | 25.9  | 25.6  | 24.9  | 24.7  | 24.8  | 25.4  | 26.0  | 26.7  | (29) |  |
|                                                                        |             | MCDB        | 29.2  | 29.5  | 29.3  | 28.4  | 27.2  | 26.7  | 26.2  | 26.2  | 26.4  | 26.9  | 27.8  | 28.5  | (30) |  |
|                                                                        | 5%          | WB          | 26.9  | 27.1  | 27.0  | 26.2  | 25.4  | 25.1  | 24.2  | 24.0  | 24.2  | 24.8  | 25.5  | 26.2  | (31) |  |
|                                                                        |             | MCDB        | 28.8  | 29.1  | 28.9  | 28.0  | 26.9  | 26.3  | 25.5  | 25.5  | 25.7  | 26.5  | 27.4  | 28.1  | (32) |  |
|                                                                        | 10%         | WB          | 26.5  | 26.7  | 26.6  | 25.8  | 25.0  | 24.5  | 23.6  | 23.5  | 23.7  | 24.3  | 25.1  | 25.9  | (33) |  |
|                                                                        |             | MCDB        | 28.4  | 28.7  | 28.5  | 27.7  | 26.6  | 25.9  | 25.1  | 25.0  | 25.3  | 26.1  | 27.0  | 27.8  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 3.8   | 3.9   | 3.7   | 3.5   | 3.2   | 3.4   | 3.6   | 3.5   | 3.8   | 3.7   | 3.6   | 3.7   | (35) |  |
|                                                                        |             | MCDBR       | 4.2   | 4.3   | 4.2   | 4.1   | 3.5   | 3.8   | 4.0   | 4.1   | 4.0   | 4.0   | 4.0   | 4.1   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 2.5   | 2.4   | 2.5   | 2.5   | 2.2   | 2.6   | 2.9   | 2.7   | 2.6   | 2.5   | 2.4   | 2.3   | (37) |  |
|                                                                        |             | MCWBR       | 3.9   | 4.1   | 3.9   | 3.7   | 3.3   | 3.5   | 3.5   | 3.7   | 3.8   | 3.7   | 3.8   | 3.8   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.387 | 0.380 | 0.378 | 0.396 | 0.380 | 0.377 | 0.366 | 0.371 | 0.396 | 0.379 | 0.380 | 0.385 | (40) |  |
|                                                                        |             | taud        | 2.613 | 2.641 | 2.656 | 2.583 | 2.610 | 2.598 | 2.622 | 2.603 | 2.528 | 2.576 | 2.600 | 2.595 | (41) |  |
|                                                                        | Ebn at Noon |             | 957   | 953   | 932   | 875   | 853   | 835   | 859   | 886   | 900   | 942   | 958   | 959   | (42) |  |
|                                                                        |             | Edh at Noon | 103   | 99    | 94    | 95    | 86    | 84    | 84    | 91    | 105   | 104   | 104   | 105   | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 6.33  | 6.08  | 5.64  | 4.70  | 4.05  | 3.65  | 3.93  | 4.51  | 5.26  | 5.99  | 6.31  | 6.55  | (44) |  |
|                                                                        | RadStd      |             | 0.61  | 0.60  | 0.42  | 0.54  | 0.26  | 0.28  | 0.24  | 0.39  | 0.40  | 0.42  | 0.50  | 0.45  | (45) |  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                        | N/A   | N/A     | +1.04  | N/A     | +0.71 | +0.78 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page